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Lecture 5: Inductive bias, Learning theory



Learning theory

- The theoretical study of
  - what is learnable?
  - how much data is needed?
  - how much work does it take?
- Studied in fields known as
  - Computational Learning Theory
  - Statistical Learning Theory
- This lecture presents a small sample from the field of learning theory, starting with the important concept of *inductive bias*

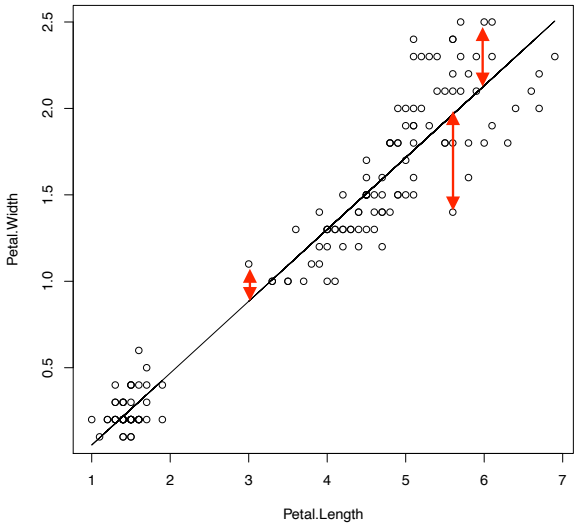
Inductive bias

Inductive bias

- **Every** learning method makes certain **assumptions** about the data
- This set of assumptions is called its inductive bias (not to be confused with statistical bias, as in bias-variance)
- We illustrate this here for linear regression and decision trees
- Important theoretical result: learning without inductive bias is impossible

Linear regression

- "Linear model" : $Y = a + b_1 X_1 + b_2 X_2 + \cdots + b_m X_m$
- Usually: best fit means minimal sum of squared vertical deviations from line ("least squares" method). (this is a choice — other loss functions could be chosen)



to predict petal *width* from petal *length* for iris flowers:

best possible linear predictor (in the sense of "least squares") is

*width = 0.416*length - 0.363*

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Interpretation of the model

What is the meaning of the b_i coefficients in $Y = a + b_1X_1 + b_2X_2 + \cdots + b_mX_m$?

- b_i tells us *how much* Y *increases* if X_i *increases by* 1 <u>and</u> *all* X_j, $j \neq i$, *remain the same*
- We call this the **effect** of X_i on Y
- Watch out: the effect is not necessarily causal!
  - If we *observe* an increase by 1 for X_i, we expect Y to increase by b_i
  - If we *make* X_i increase by 1, we do not know what will happen to Y
  - (E.g.: predicting outside temperature from people's clothing. If people dress lightly, it's probably warm, but you can't make it warmer by dressing lightly)

Important assumptions

- The linear regression model implicitly assumes:
  - The effect of each variable on the target is **constant** (does not depend on other variables) — indeed, if it's not constant, the interpretation of the b_i mentioned before would be meaningless
  - The effects of different variables are **cumulative** (add up)
  - (there are some more technical assumptions, which do not matter here)

- In statistics terminology: "**No interaction**"

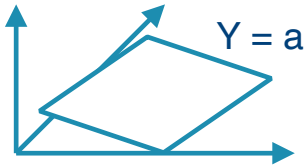
Decision tree induction

- After splitting on X_i, each branch is developed entirely independently
  - X_j may have a positive effect on Y in one branch, but a negative in another branch
  - No assumption whatsoever that "X_j has the same effect everywhere"
- From this point of view, decision trees are quite the opposite of linear regression !

Trees vs. linear regression

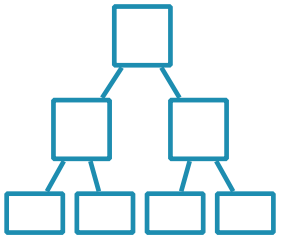
A,S,C,J: Age, Sex, Country, Job

$$Y = a + b_1A + b_2S + b_2C + b_3J$$



On average, professor salaries are 500$ higher than postdoc salaries. I'll assume the same holds for each country separately.

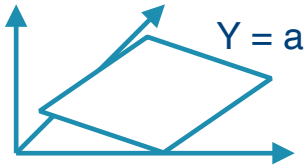
Nah… I don't think so. Even if very strong evidence to the contrary is presented, I'll stick to this assumption.



That's silly! You don't know that. It may be different in different countries.

Wow… you're quite biased!

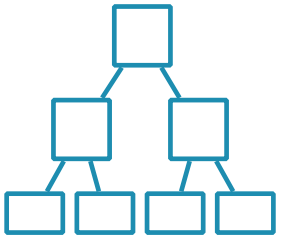
Trees vs. linear regression



$$Y = a + b_1A + b_2S + b_2C + b_3J$$

Well, surely that'll be the same, no?

You're quite paranoid!



Professors earn 500$ more than postdocs if we look at males aged 30-40; males aged 40-50; males aged 50-60; females aged 30-40; females aged 50-60. But what about females aged 40-50? I have no data!

I'll believe that when I see it with my own eyes, not before.

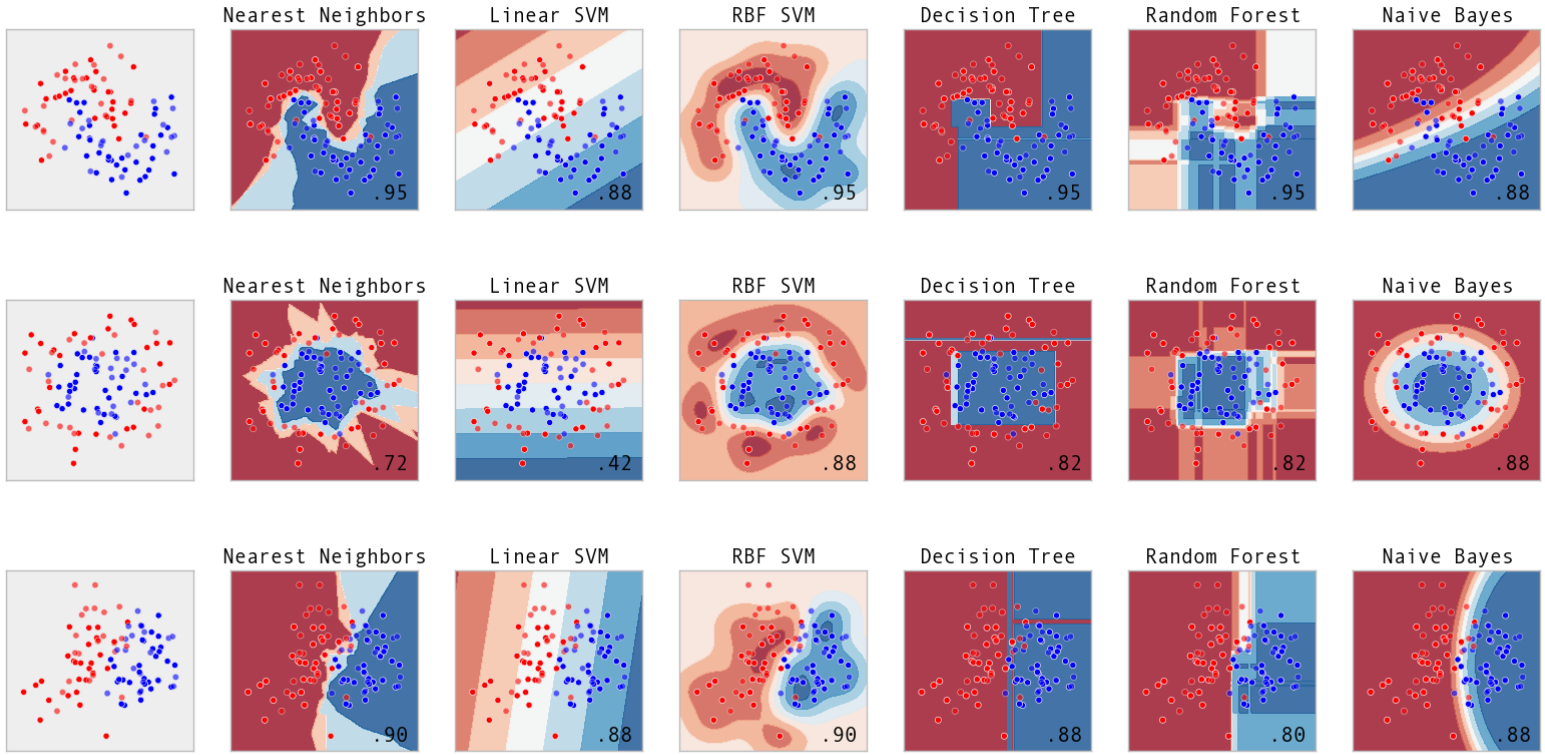
Trees vs. linear regression

- Which method performs best? Largely depends on your problem!
- Each learning approach has an “inductive bias”: implicit assumptions it makes
- Learners whose bias fits your problem will perform better
- Unfortunately, the problem is often not sufficiently understood to decide in a principled way (neither are learners, in fact!)
- “Cumulative effects” vs. “interaction” is just one simple example of how methods can be biased

Many methods, many biases

- Many different machine learning methods exist
- They can be used for all kinds of modeling, including predictive modeling
- Examples: Naive Bayes, probabilistic graphical models, discriminant analysis, …
- All of them have their own bias = implicit assumptions about what properties the true model has
- This results in different ways of generalizing towards unseen data
- The next slide illustrates this for classification

Classifiers illustrated in 2-D



(source code: Scikit-Learn documentation)

Get rid of bias?

- Theoretical proofs exist that **bias-free learning is impossible**
- Wolpert's "no free lunch theorems": For each problem where method A works better than B, there's a problem where B works better than A (using a probabilistic characterization). *There is no single best method for learning.*
- A less technical argument (presented by Mitchell, 1996):
  - Let hypothesis space $\mathcal{H}$ = set of all hypotheses that the learner will consider, and *version space* VS = subset of $\mathcal{H}$ consistent with the training set
  - What happens if $\mathcal{H}$ contains every possible concept (i.e., every imaginable function from X to {True,False}?
  - Then for any hypothesis $h \in VS$ that predicts True for a new instance q, there exists a $h' \in VS$ that predicts False for q (just choose $h'(x) = h(x)$ for all $x \neq q$ and $h'(q) = \neg h(q)$).
  - Hence, if we let all $h \in VS$ vote, the votes will be exactly 50-50 for every unseen instance
- General conclusion: **no generalization towards unseen cases is possible without making some assumptions**

Bias

- Bias can be in the form of preference or selection bias
  - Selection bias: not every imaginable hypothesis is in the hypothesis space
  - Preference bias: some hypotheses in the hypothesis space are preferred over others
    - e.g., decision trees: preference for small trees over large ones
- In the remainder, we will focus on selection bias
- The closer a hypothesis space is to being able to represent every imaginable hypothesis, the smaller the inductive bias is
- *How can we measure the strength of the bias, and how does it impact learnability?*

Measuring the expressiveness, or “capacity”, of a hypothesis space

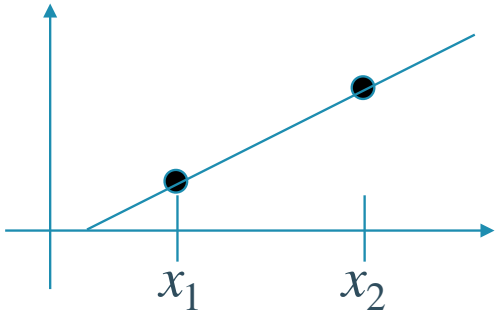


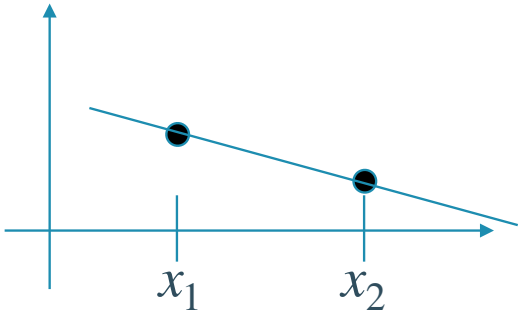
Capacity of a hypothesis space

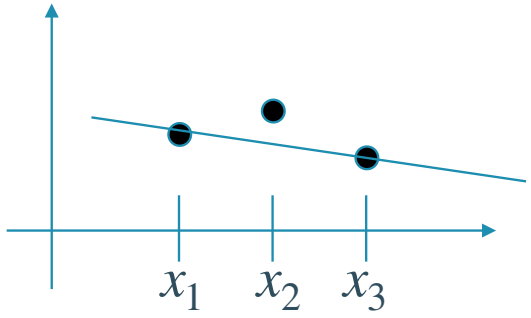
- Hypothesis space $\mathcal{H}$: set of all models ("hypotheses") that could be returned by the learner
- We will here focus on predictive models (functions $f : \mathcal{X} \rightarrow \mathcal{Y}$)
- Question: How expressive / flexible is a given hypothesis space? Can it model a wide variety of functions / concepts, or a rather limited set?
- One way to answer this question is the following: given a dataset D, can we always find a model f that perfectly fits it (i.e., $f(\boldsymbol{x}) = \boldsymbol{y}$ for all $(\boldsymbol{x}, \boldsymbol{y}) \in D$)? How large should D be before this is no longer the case?
  - This expresses $\mathcal{H}$'s "capacity to perfectly fit" data
  - Note: this also implies a capacity to *overfit* data
  - We simply call it the **capacity** of the hypothesis space

Example: polynomial regression

- Consider univariate regression: we look for functions of the form $y = ax + b$
- Assume two inputs x_1 and x_2 are given. You get to choose the labels y_1 and y_2, I have to find values for a and b such that $y_1 = ax_1 + b$ and $y_2 = ax_2 + b$
- No matter how you choose y_1 and y_2, I can always find the right values for a and b. I can always find a line that goes through both points.
- For *three* points, however, this is no longer true: it is quite easy to find y-values for the x_i such that no straight line connects all three points

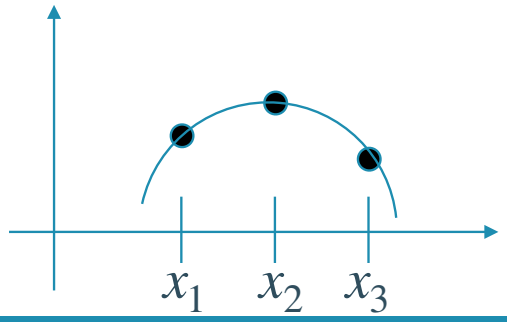






Example: polynomial regression

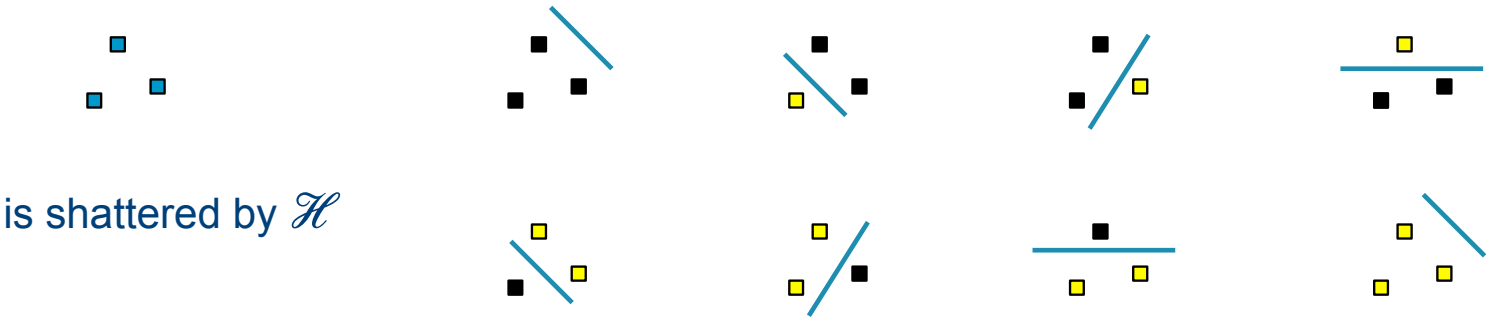
- However, if we consider *quadratic* functions (of the form $y = ax^2 + bx + c$), then it is possible to perfectly fit three points
- Generally, polynomials of degree n can fit $n + 1$ points, in the sense that no matter how you choose the y-values, a polynomial exists that perfectly fits the data
- This means the hypothesis space "polynomials of degree n" is more expressive (has a higher capacity) than the hypothesis space "polynomials of degree $n - 1$"



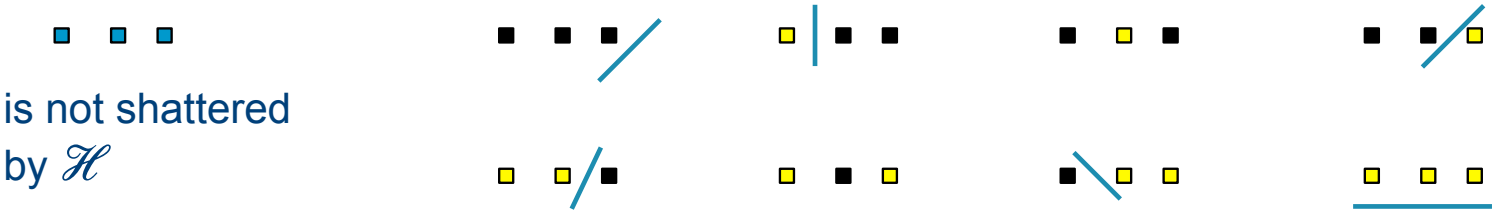
The principle of capacity, applied to binary classification

- We will now apply the same principle to classification
- Consider binary classification: $y \in \{0,1\}$
- **Definition**: A set S of n points is **shattered** by a hypothesis space if, for *each* possible label assignment $l : S \to \{0,1\}$ to these points, a hypothesis $f \in \mathcal{H}$ exists such that $\forall \boldsymbol{x} \in S : f(\boldsymbol{x}) = l(\boldsymbol{x})$
- That is: given $S = \{x_1, x_2, \ldots, x_n\}$, no matter how you choose the corresponding labels y_i, it is always possible to find a hypothesis that perfectly fits the data (similar to the regression case)

Example: linear separators in 2D



is shattered by $\mathscr{H}$



is not shattered by $\mathscr{H}$

labels

Hypothesis

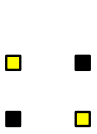
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VC-dimension

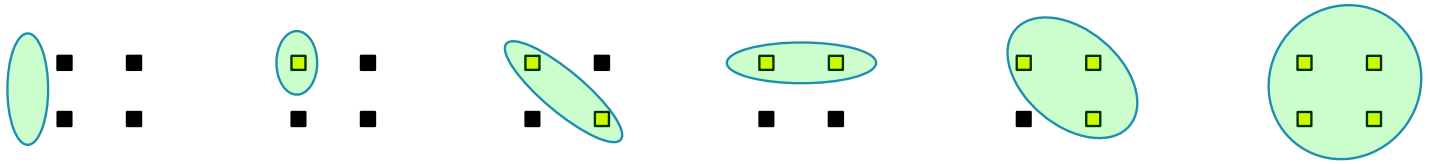
- **Definition**: The VC-dimension (or Vapnik-Chervonenkis dimension) of a hypothesis space $\mathcal{H}$ containing functions $\mathcal{X} \rightarrow \{0,1\}$ is the size of the largest finite subset of $\mathcal{X}$ that is shattered by $\mathcal{H}$, or infinite if shattered subsets exists for any size.
- Note: not all subsets of size k have to be shattered. It suffices that one shattered subset of size k exists for the VC-dimension to be at least k.

Example: 4 points in 2D

- The VC-dimension of linear separators in 2D is 3 : there does not exist a set of 4 points that is shattered by this hypothesis space (we omit the proof)
- With hypotheses of the form “everything inside the ellipse is positive” (where the ellipse can be chosen), there exists a set of 4 points that is shattered — so the VC-dimension of this hypothesis space is at least 4



No linear separator separates □ from ■



(10 more labelings exist, but all equivalent to one of the above)

An upper bound on the VC dimension

- Note: a set of n points can be labeled in 2^n ways
- For this set to be shattered by $\mathscr{H}$, at least one hypothesis in $\mathscr{H}$ must exist for each labeling, so $\mathscr{H}$ must contain at least 2^n hypotheses
- In other words, $VC(\mathscr{H}) = n$ implies $|\mathscr{H}| \geq 2^n$
- Written differently, $|\mathscr{H}| \geq 2^{VC(\mathscr{H})}$, and therefore, $VC(\mathscr{H}) \leq \log_2(|\mathscr{H}|)$

PAC-learning

How much data is needed for successful learning?

Sample complexity & its relation to capacity

Statistical learning theory

Statistical learning theory provides theoretical results that relate:

- probability of “successful” learning
- number of training examples (**sample complexity**)
- how training examples are chosen
- complexity of hypothesis space
- accuracy to which target is approximated
- time spent on learning (**time complexity**)

Complexity of concept learning

We will focus on the following task (binary classification / concept learning) :

- given:
  - instance space $\mathscr{X}$
  - existence of an unknown target function $c : \mathscr{X} \to \{0,1\}$
  - hypothesis space $\mathscr{H}$
  - training examples $D \subseteq \mathscr{X} \times \{0,1\}$
- find:
  - a function $f \in \mathscr{H}$ such that $f(x) = c(x)$
    - for all $x \in D$?
    - for all $x \in \mathscr{X}$? (this is most interesting)

PAC learning

- PAC stands for “probably approximately correct”
- Assume examples are drawn randomly from one distribution (***i.i.d.*** : independently and identically distributed)
- We can never *guarantee* that a correct hypothesis is learned: there’s always a small chance that randomly drawn examples are highly unrepresentative
- Therefore, relax the learning task: learner should *very probably* find an *approximately correct* hypothesis
- Formally: with probability $\geq 1 - \delta$, return a hypothesis with accuracy $\geq 1 - \varepsilon$
- The smaller δ and ε are chosen, the stronger this “soft guarantee” is

PAC-learnability

- Formal definition of PAC-learnability according to Mitchell (1997):
  - Assume a class C of possible target concepts defined over instance space X; instances have "size" n; learner L uses hypothesis space $\mathcal{H}$
  - C is **PAC-learnable by L using $\mathcal{H}$** if for all c in C, for all probability distributions over X, and for all $0<\delta<1/2$ and $0<\varepsilon<1/2$, the following holds: L finds with probability $1-\delta$ some $f \in \mathcal{H}$ with acc(f) ≥ $1-\varepsilon$ in time polynomial in $1/\delta$, $1/\varepsilon$, n and size(c)

Linking training error and real error

- Consider binary prediction, learning from an i.i.d. dataset
- Can we (probabilistically) bound the error on the population, based on the error in the training set and the size of the training set?
- Many results on this exist, often complicated to derive
- To illustrate the type of reasoning, we look at a simple case here:
  - If I have a hypothesis that perfectly fits a training set of size n… (error 0)
  - … how likely is it that its real error is still larger than some ε ?

ε-exhausting the version space

- When the version space (the set of all hypotheses that perfectly fit the training set) does not contain any hypotheses with error $\geq \varepsilon$, we say that the version space is **ε-exhausted**
- We will try to answer the following question: given values for δ, ε, how many instances does the dataset have to contain so that, with probability at least $1 - \delta$, the version space is ε-exhausted ?

Haussler's theorem (1988)

- Assume hypothesis h has real error ε. How likely is it that it perfectly fits a data set of n randomly drawn instances? That probability is $(1-\varepsilon)^n$
- It h has error $\geq \varepsilon$, this probability is $\leq (1-\varepsilon)^n$
- If there are k such hypotheses, the probability that at least one perfectly fits the dataset is $\leq k(1-\varepsilon)^n$
- Clearly, $k \leq |\mathcal{H}|$ (you can't have more such hypotheses than there are in the entire hypothesis space)
- Hence, the probability that at least one hypothesis with error $\geq \varepsilon$ perfectly fits the data is upper-bounded by $|\mathcal{H}|(1-\varepsilon)^n$
- $|\mathcal{H}|(1-\varepsilon)^n$ is itself upper-bounded by $|\mathcal{H}|e^{-n\varepsilon}$
- Hence: if $h \in VS$, then $P(\text{error}(h) > \varepsilon) \leq |\mathcal{H}|e^{-n\varepsilon}$ (Haussler's theorem)

How many examples needed?

- From the previous slide, it follows that $|\mathcal{H}|e^{-n\varepsilon} \leq \delta$ guarantees that the probability that the version space contains at least one hypothesis with error $\geq \varepsilon$ is at most δ
- Solve for n: $n \geq \ln(|\mathcal{H}|/\delta)/\varepsilon$; or, equivalently, $n \geq \frac{1}{\varepsilon}\left(\ln|\mathcal{H}| + \ln\frac{1}{\delta}\right)$
- Conclusion: learning from at least $\ln(|\mathcal{H}|/\delta)/\varepsilon$ instances guarantees with probability at least $1-\delta$ that
  - the version space contains only ε-correct hypotheses
  - Hence, any learner L that returns a perfectly-fitting classifier, returns an ε-correct hypothesis

A concrete example

- Consider learning a conjunctive formula over 100 boolean variables. A variable may occur in the conjunction positively, negatively, or not at all. If you want to be 99% certain that the learned formula gives a different result than the real one on at most 1% of the population, how many examples should you use?

- Answer: $|\mathscr{H}| = 3^{100}$ (for each of the 100 variables, there is the option to include it positively or negatively, or leave it out from the formula). We thus obtain $n \geq \ln(3^{100}/0.01)/0.01 \approx 11447$. (The total instance space has size $2^{100} = 1.268 \cdot 10^{30}$)

Sample complexity and VC-dimension

- In the previous examples, $|\mathcal{H}|$ served as a measure for how expressive $\mathcal{H}$ is (the more hypotheses in $\mathcal{H}$, the more different concepts it can represent)
- What if $|\mathcal{H}|$ is infinite? (e.g.: linear regression, neural nets) The result is useless in that case!
- Also: perhaps many $f \in \mathcal{H}$ are actually quite similar; to what extent does $\mathcal{H}$ contain truly different hypotheses?
  - E.g., $\mathcal{H}_1 = \{y = ax + b \,|\, a, b \in \mathbb{R}\}$ vs. $\mathcal{H}_2 = \{y = ax^2 + bx + c \,|\, a, b, c \in \mathbb{R}\}$: one is more expressive than the other
- This is where capacity measures such as the VC-dimension come in

Sample complexity and VC-dimension

- Many results exist that link learnability to VC-dimension
- E.g.: how many examples needed to ε-exhaust VS with probability at least 1-δ ?

$$n \geq \frac{1}{\varepsilon}\left(4\log_2\frac{2}{\delta} + 8VC(H)\log_2\frac{13}{\varepsilon}\right)$$

- Under mild conditions,

$\mathscr{H}$ is PAC-learnable ⇔ $VC(\mathscr{H})$ is finite

- Many known results are based on theory about Rademacher complexity — which we will not go into here.

Summary

- Sample complexity relates the chances of successful learning to the size of the training set
- The sample complexity of a learning problem is related to the expressiveness of the hypothesis space
- The latter can be formally expressed by measures such as the size of the hypothesis space (for finite sizes), its VC-dimension, and its Rademacher complexity

Some final remarks…

- PAC-learning assumes *random sampling*. It does not say anything about learning settings where the instances are carefully chosen by a benevolent teacher, or by the learner, to be maximally informative (e.g., the *active learning* setting)
- Sample complexity is only one measure. Computational complexity is another possibly useful measure: how much *computation* is needed, rather than how much data